MUHAMMAD ILHAM

Software Developer | AI Application Development | DevOps Engineer
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PROFILE SUMMARY

Software Developer with hands-on experience in web application development, API-driven backend services, and AI integration. Experienced with React, Next.js, TypeScript, Python, and FastAPI, with additional experience in Laravel and modern DevOps technologies such as Linux, Docker, Kubernetes, and Nginx. Interested in building reliable and scalable software solutions while continuously developing expertise in AI application development, cloud computing, infrastructure engineering, and automation.

TECHNICAL SKILLS

Infrastructure and Operations: Kubernetes (multi-node, Nginx Ingress), Docker, Proxmox Virtual Environment (LXC/KVM), Linux server administration, AWS EC2, Microsoft Azure Virtual Machines, containerd, kubeadm, kubectl, kubelet, NFS storage, Vercel
Networking and Security: Nginx reverse proxy and caching, ModSecurity WAF (OWASP CRS), TLS termination, load balancing, Bash scripting, Apache JMeter (load testing)
Software Development: TypeScript, JavaScript, React, Next.js (SSR, SSG, App Router), HTML, CSS, Tailwind CSS, Laravel, Livewire, Node.js (basic), FastAPI (basic), REST API design, authentication and role-based access control (RBAC)
Database, Tools and Quality Assurance: PostgreSQL, SQLite, Prisma ORM, DBMS, Git/GitHub, Lighthouse (performance and SEO auditing)

WORK EXPERIENCE

DevOps Intern – PT. cmlabs, Malang, East Java Aug 2025 – Dec 2025
Development Division (SEO-focused company)

- Built an SEO-friendly internal landing page using Next.js (App Router), implementing 7 page groups (Homepage, About Us, Our Team, Services, Portfolio, Blog, and Contact Us) based on the company's Figma design system with reusable React components.
- Implemented technical SEO optimizations, including page-specific metadata, Open Graph/Twitter cards, canonical URLs, sitemap, robots.txt, and image optimization using next/image.
- Managed application deployment through Vercel's GitHub integration, with automatic builds triggered by Git pushes and manual deployment when needed, achieving a good Lighthouse SEO score.

PROJECTS

Waves – Multi-Function Web Toolkit Sep 2026 – Present
Associated with Politeknik Negeri Jember kiraryuuki.github.io/Waves
Developed With: ***TypeScript, FastAPI, Python***

- Designed and built a local-first toolkit with nine utilities, including stem separation (Demucs), voice synthesis and fine-tuning (VITS), image and video generation, a GGUF/Ollama LLM hub, and a background remover, on a Vite and FastAPI stack.

PANDAWA – Geospatial Natural Resources Information System Feb 2025 – Jun 2025
Associated with Politeknik Negeri Jember kiraryuuki.github.io/PANDAWA
Developed With: ***TypeScript, D3.js, Next.js, Prisma, PostgreSQL***

- Built the web platform from scratch to release for monitoring natural-resource data in Bondowoso Regency: interactive map, district detail pages, dashboard, and harvest-yield prediction.
- Co-creator of the registered software copyright No. 000942745 (Jul 2025, Directorate General of Intellectual Property).

Amfibi – Car Marketplace and Rental Platform Feb 2024 – Jun 2024
Associated with Politeknik Negeri Jember github.com/KiRaRyuuKi/Amfibi
Developed With: ***Laravel, Livewire, Tailwind CSS, PostgreSQL***

- Designed and built an end-to-end multi-showroom marketplace for new and used cars with web-based rental features, including real-time chat, rental-availability calendar, booking confirmation, and automated invoicing.
- Implemented Laravel Livewire components for purchase, rental, and chat interfaces; integrated Laravel Echo and Pusher for real-time communication; implemented role-based access control (RBAC) for admin, seller, and customer roles; and migrated the database from MySQL to PostgreSQL.

RabHBiTs – Pakcoy Leaf Rot Detection System Aug 2024 – Dec 2024
Associated with Politeknik Negeri Jember github.com/KiRaRyuuKi/RabHBiTs
Developed With: ***Flutter, Python, scikit-learn***

- Developed a machine learning model using K-Nearest Neighbors (KNN) to classify pakcoy leaves as healthy or rotten based on image data.
- Handled the end-to-end development process, including dataset collection, model training, and integration into a Flutter mobile application.

RisqiShop – Cashier and Inventory Management Feb 2023 – Jun 2023
Associated with Politeknik Negeri Jember github.com/KiRaRyuuKi/RisqiShop
Developed With: ***Java, SQLite***

- Developed a desktop app for sales recording with automatic discount and total calculation, stock management, and daily/monthly reports exported to PDF and Excel; migrated the database from MySQL to SQLite.

EDUCATION

D4 Informatics Engineering 2022 – 2026
Politeknik Negeri Jember – Jember, East Java

Computer and Network Engineering 2019 – 2022
SMK Negeri 1 Bondowoso – Bondowoso, East Java

THESIS

ModSecurity and Nginx Caching on Kubernetes Nov 2025 – Jun 2026
Politeknik Negeri Jember – Jember, East Java sipora.polije.ac.id/id/eprint/57452
Kubernetes, ModSecurity, Proxmox, Nginx, JMeter

- Designed a dual-layer defense on Nginx Ingress Controller in a multi-node Kubernetes cluster (Proxmox Virtual Environment on AWS EC2, 2 vCPU, 8 GB RAM) and load-tested it with Apache JMeter on WordPress, Ghost, and Drupal at up to 1,000 concurrent users.
- Blocked 95% of HTTP(S) flood traffic and held app-pod CPU under 25 millicores with the combined mitigation. Documented the results in a 160-page thesis.

CERTIFICATIONS

BNSP Web Developer – LSP Politeknik Negeri Jember 2026
BNSP Network Administrator (KKNI Level II) – LSP SMK Negeri 1 Bondowoso 2022
Claude Code in Action and Introduction to Claude Cowork – Anthropic 2026

ACHIEVEMENTS

Copyright Registration – PANDAWA (Pantau Sumber Daya Alam Wilayah Bondowoso) 2025
Recognized as one of the creators of PANDAWA, a geospatial information system developed for mapping, monitoring, and analyzing natural resources in Bondowoso Regency using real-time data to support decision-making. Officially registered with the Directorate General of Intellectual Property (DJKI), Ministry of Law of the Republic of Indonesia, under Copyright Registration No. 000942745.

LANGUAGES

Indonesian - Native (professional)